Literature Review for Master Thesis

1 Nomenclature

Symbols, definition and unit

$\boldsymbol{t_o}$: Subfault onset time [s], e.g. $t_o = (x - x_{\text{start}})/v_{\text{rupture}}$.

$\boldsymbol{x}$, $\boldsymbol{x}_{\text{start}}$: Horizontal coordinate and rupture starting position [m].

$\boldsymbol{v}_{\text{rupture}}$: Prescribed rupture propagation speed [m s^{-1}].

$\boldsymbol{t}_{rise}$: Rise time of a subfault/source [s].

$\boldsymbol{\Delta x}$: Subfault width or discretisation length in x [m].

$\boldsymbol{x}_{\text{slip}}$: Prescribed slip on a subfault [m].

$\boldsymbol{M}_0$: Scalar seismic moment per subfault [N m].

$\boldsymbol{m}_{ij}$: Moment tensor components [N m].

$\boldsymbol{r}_{\text{mode}}$: Mode mix parameter [dimensionless].

$\boldsymbol{l}_{cz}$: Cohesive process (fracture) zone length [m].

$\boldsymbol{\tau}_s$: Shear strength of interface [Pa].

$\boldsymbol{l}_{ch}$: Irwin characteristic length [m].

$\boldsymbol{\lambda}_{\min}$: Shortest wavelength to resolve [m].

$\boldsymbol{v}_{\min}$, $\boldsymbol{v}_{\max}$: Minimum and maximum relevant wave speeds in the model [m s^{-1}].

$\boldsymbol{\Delta t}$: Time step [s] for explicit time integration.

$\boldsymbol{\xi}$: Normalized opening [dimensionless].

$\boldsymbol{\delta}$: Local opening (separation) displacement across the interface [m].

$\boldsymbol{\delta}_c$: Mixed-mode critical opening [m] used to nondimensionalize openings.

$\boldsymbol{\delta}_{n,c}$, $\boldsymbol{\delta}_{s,c}$: Single-mode critical openings for Mode I (normal) and Mode II (shear) [m].

$\boldsymbol{T}(\boldsymbol{\xi})$: Cohesive traction as a function of normalised opening [Pa or N m^{-2}].

$\boldsymbol{T}_{\text{peak}}$: Peak cohesive traction [Pa].

$\boldsymbol{\phi}(\boldsymbol{\Delta})$: Mixed-mode cohesive potential [J m^{-2}] with $\Delta = (\Delta_n, \Delta_t)$.

$\boldsymbol{\Delta}_n$, $\boldsymbol{\Delta}_t$: Normal and tangential separations [m].

$\boldsymbol{\delta}_n$, $\boldsymbol{\delta}_t$: Normal and tangential length scales in the cohesive potential [m].

$\boldsymbol{q}$: Coupling parameter in the mixed-mode cohesive potential [dimensionless].

$\boldsymbol{\phi}_n$, $\boldsymbol{\phi}_t$: Work (energies) of separation for pure normal and tangential modes [J m^{-2}].

$\boldsymbol{\sigma}_{\max}$, $\boldsymbol{\tau}_{\max}$: Peak normal strength and peak shear strength [Pa].

$\boldsymbol{\sigma}_n$, $\boldsymbol{\tau}_s$: Normal and shear strength [Pa].

$\boldsymbol{G}_I$, $\boldsymbol{G}_{II}$: Fracture energies for Mode I and Mode II [J m^{-2}].

$\boldsymbol{v}_s$, $\boldsymbol{v}_p$: Pressure and Shear seismic wave speeds [m s^{-1}].

$\boldsymbol{\rho}$: Mass density [kg m^{-3}].

$\boldsymbol{\mu}$, $\boldsymbol{\lambda}$: Lamé parameters [Pa].

$\boldsymbol{E}$: Young's modulus [Pa].

$\boldsymbol{l}_{crack}$, $\boldsymbol{x}_0$: Local crack process length and initial crack front position [m] used in local coordinate definitions.

$\boldsymbol{\nabla}$: Component-wise partial derivative operator (nabla).

2 Introduction for Thesis

Snow avalanches are a major natural hazard in mountainous regions; they endanger human life as well as infrastructure (Schweizer et al., 2003). The Swiss Institute for Snow and Avalanche Research has reported an average of 22 fatalities caused by avalanches every year, with the number of non-fatal accidents being tenfold higher (SLF, 2025). These data show that avalanches are a significhant threat, which is why it is crucial to understand how avalanches form and propagate in order to protect people and infrastructure better.

Avalanches are rapid masses of snow that move down mountain slopes due to gravity (Ancey, 2001). Snow avalanches can either be loose snow, which starts from a point in a cohesionless surface, wet snow, or snow slab avalanches (Schweizer et al., 2003). In this Master Thesis, the focus will be on dry snow slab avalanches, so mostly coherent masses of snow which contain no liquid water. Snow slab avalanches consist of a cohesive slab of snow (Schweizer et al., 2003) that is released because of loading at the surface, which triggers crack propagation in a weak layer (Gaume et al., 2017). Weak layers are characterized by the grain shape of snow, which leads to lower cohesion of those layers (Föhn et al., 1998).

As mentioned, avalanches are a major threat to human life in mountainous regions, which is why detecting the breakoff of the snow slab is important for hazard mitigation. Detecting the breakoff of the slab means that warnings can be issued in lower parts of the mountain that an avalanche is possible. Furthermore, detecting slab breakoff with DAS is also important to further understand crack dynamics in snow as well as the factors influencing crack propagation.

In this master's thesis, the aim is to detect avalanche slab breakoff and crack propagation using modeled data from distributed acoustic sensing (DAS) cables.

In this thesis, a fibre-optic cable installed on the slope is treated as a continuous seismic sensor, so that deformation can be observed along its full length rather than at only a few isolated stations (Parker et al., 2014). This is possible because DAS sends optical pulses through the fibre and records the backscattered light from every position along the cable (Shatalin et al., 1998). Ground motion near the cable, for example weak-layer failure or slab breakoff, changes the fibre strain, which appears as phase variations in the backscattered signal (Hartog, 2017). The phase variations are then converted into dynamic strain and from strain, into deformation along the cable (Trabattoni et al., 2023). mention also the cable location here, length etc

A pilot early-warning system using distributed acoustic sensing has been implemented in Norway, where avalanches are detected by their seismic signature (Turquet et al., 2024), meaning when the avalanche is already moving downhill. Such sensors are also located in Vallée de la Sionne in the Swiss Alps (Paitz et al., 2023). The detectability of avalanches by DAS has also been studied in the Swiss Alps by Paitz et al., 2023 and Edme et al., 2023, but the detectability of the slab breakoff itself has not been studied.

A multitude of models are used to study different processes in snow. The SNOWPACK and CROCUS models used in the Swiss and French alps model the structure and evolution of snow (Bartelt & Lehning, 2002; Brun et al., 1992). These models were developed to understand how snow evolves under different conditions, for example a changing temperature gradient and water content and what consequences this has for the formation of weak layers where failure likely occurs and avalanches are triggered (Bartelt &

Lehning, 2002; Brun et al., 1992). SNOWPACK and CROCUS are comprehensive models that take into account multiple governing equations and processes and what consequences they have for the structure of snow. While it is important to understand the influences on for example temperature gradient on the snow structure and weak layer formation and stability as a consequence, this is not the focus of the thesis. This is because these models are too complex and comprehensive for the purpose of trying to detect slab breakoff in DAS data as they model the entire snow structure. Using SNOWPACK or CROCUS to investigate how different snowpack compositions and factors influence the DAS signal is beyond the scope of this thesis.

To model crack propagation in snow, multiple numerical modeling methods have been used. Cundall and Strack, 1979 developed the distinct element method (DEM) for granular media. This model discretizes the domain as spheres and disks, which interact with each other based on Newton's second law (Cundall & Strack, 1979). The interactions of particles are modeled by the governing equations and the resulting forces and displacements (Cundall & Strack, 1979). This method has been used by Bobillier et al., 2020, 2024; Mulak and Gaume, 2019 and many others to model crack propagation and its influencing factors. The main limitation of DEM is that the particle contacts and the corresponding equations need to be recalculated every time a grain contact is made or broken (Cundall & Strack, 1979). This means that for large, slope-scale simulations, this method becomes increasingly computationally expensive. In addition to the high computation cost, DEM also cannot account for anticrack propagation, the closure of cracks in this propagation mode is accounted for by manually removing elements from the mesh, which is not very efficient (Gaume et al., 2019).

This problem is solved by using the material point method (MPM). In this method, the material is discretized in a set of Lagrangian points, which are used to track mass, deformation and momentum (Bardenhagen et al., 2000). Because the points are not fixed, an Eulerian background mesh is needed to calculate spatial derivatives (Bardenhagen et al., 2000). This method is not only more computationally efficient on the slope-scale, it can also account for anticrack behavior and strain softening of snow (Gaume et al., 2019). This is why the material point method will also be used in the thesis to investigate slab breakoff in DAS data on a large scale. The material point method has been used by Bergfeld et al., 2025, who modeled slope-scale experiments to understand the transition of crack propagation modes, Trottet et al., 2022 who modeled propagation saw tests to observe transitions in crack propagation speeds, as well as Gaume et al., 2019 and Gaume et al., 2018.

While it is important to understand the different modeling methods and their limitations, it is also important to understand the different types of crack propagation and what factors influence them. Understanding how fast a crack can propagate and which factors influence this is important for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

Two main types of crack propagation are believed to be important for weak layer failure: anticrack propagation and shear crack propagation (Bergfeld et al., 2025). There are three ways in which cracks can open up: mode I, where space is opened between the crack faces under tensile loading, and modes II and III, which represent parallel shearing and diagonal shearing (Heierli et al., 2003). In anticrack propagation, the weak layer consolidates under loading (Knight & Knight, 1972), while cracks propagate in a combination of modes I-III (Heierli et al., 2003). Gaume et al., 2018 modeled anticrack propagation induced by a propagation saw test (PST) using MPM, which showed that the scale of the simulations influences the propagation direction of the cracks. For smaller scale simulations, cracks propagate from top to bottom of a slope, whereas for slope-scale simulations, cracks propagate bottom to top, which means that remote triggering of avalanches is possible (Gaume et al., 2018). This is an important consideration for the thesis because it influences where the slab breakoff would be observed in terms of DAS cable location. Other models also show that the crack propagation speed of anticracks is influenced by slope angle and curvature: concave slope areas have higher propagation speeds than convex areas of the slope (Gaume et al., 2019). Furthermore, there is a directionality in propagation speed, simulations by Gaume et al., 2019 show that cracks propagate twice as fast upslope than in the downslope direction, which is especially important when thinking about remote triggering. The slab breaks off when the slope angle exceeds the friction angle of the material (Gaume et al., 2019). Slope angle also influences which mode of crack propagation is more dominant (Rosendahl & Weißgraeber, 2020). In steeper terrains, shear modes (II and III) dominate, whereas in flatter terrain, mode I dominates the crack propagation

(Rosendahl & Weißgraeber, 2020). The critical force, which is for example exerted by a skier, needed for weak layer failure is also dependent on slope angle; it decreases with increasing slope angle (Rosendahl & Weißgraeber, 2020). Snow parameters also have an influence on this critical force, as shown by simulations by Rosendahl and Weißgraeber, 2020: the thicker the slab or the higher the slab tensile strength, the better the force by the skier can be distributed on the weak layer and the more force is needed for failure.

Multiple experiments and numerical models have shown that cracks can propagate faster than the elastic wave speed in snow, so-called supershear crack propagation (Bergfeld et al., 2025; Bobillier et al., 2024; Trottet et al., 2022). Understanding what causes the transition from sub-Rayleigh to supershear crack propagation is of high importance for the thesis, because it can give an estimate of whether supershear crack propagation will occur under certain conditions, which leads to faster propagation of cracks and less time between loading and slab breakoff. In MPM models of PST, Trottet et al., 2022 observe that the occurrence of supershear cracks is dependent on slope angle: in steeper terrains, supershear propagation was observed. This can be explained by slab bending, because in steep terrains, bending of the slab is not limited to the collapse depth of the weak layer (Trottet et al., 2022). In in steeper terrain, the slab can bend more because of the terrain angle, the additional bending increases tensile stress in the slab, which leads to faster crack propagation (Trottet et al., 2022).These observations were also made by Bobillier et al., 2024 as well as Bergfeld et al., 2025. The release area of an avalanche is also tied to propagation speed: at supershear propagation speed, there is less crack arrest and therefore larger release areas (Bergfeld et al., 2025; Trottet et al., 2022). This is crucial because larger release areas also mean a larger snow volume, which has more potential for damage.

Independent of crack propagation mode, different factors influence the speed and distance at which cracks can propagate in the weak layer before failure occurs (Gaume et al., 2015, 2018; Gaume et al., 2017). As discussed above, slope angle and stiffness are important factors. Other factors include weak layer parameters: the weaker the weak layer, the slower crack propagation; the thicker the weak layer, the lower the crack propagation distance before failure (Gaume et al., 2015). The depth of the slab also has an influence on propagation distance and speed: the thicker the slab, the larger the propagation distance and velocity (Gaume et al., 2015). How fast a slab will break off is also dependent on the critical crack length needed for failure (Gaume et al., 2017). If the snowpack is more stable, which is the case when the slab has a high elastic modulus or the weak layer is thicker, the critical crack length is higher (Gaume et al., 2017) because the snow can sustain longer cracks before it fails. A very important factor in material strength is loading rate: if loading is slow, sintering can occur (Mulak & Gaume, 2019). This means that new snow contacts can form, which can strengthen the material (Mulak & Gaume, 2019). This is in contrast to rapid loading, which happens for example due to a skier (Mulak & Gaume, 2019).

In order to understand how crack propagation and slab breakoff would look in DAS data, numerical modeling of the wave propagation is done. This is done in SALVUS, which is a package that uses spectral elements to model seismic wavefields (Afanasiev et al., 2017).

The different snow processes described have differenit implications for the seismic wavefield, which will be modeled. Firstly, the loading which occurs before crack nucleation changes the stress state of the snowpack. The weak layer collapse and anticrack propagation produce volumetric compaction in the snowpack, this produces a seismic signal. The crack propagation, whether sub-Rayleigh or supershear, creates a moving rupture front, which can be likened to a moving source of seismic waves, the cracks itself also create a seismic signal. Lastly, the slab breakoff and sliding creates a seismic signal because a large mass of snow moves downhill, which likely creates the largest seimsic signal of all of the processes mentioned. The signals of the different processes described will be modeled in the thesis, which will help to understand how the different processes look in DAS data and whether they can be distinguished from each other.

Based on the above information, there are multiple considerations necessary in order to model DAS data of crack propagation accurately. The scale of the models is an important factor, because as mentioned, it influences the direction of crack propagation. This influences at which location along or relative to the DAS cable slab breakoff will occur, which changes the measured data. Furthermore, multiple parameters influence the time passed between initial cracking and slab breakoff, these are important to consider when interpreting data. The different modes of crack propagation are also important to consider, it is possible that a transition between sub-Rayleigh and supershear crack propagation can be observed in the data. The loading rate can also be considered, because slower loading leads to stabilizing processes such as sintering.

here i elaborate the research questions, but I don't know which one I will focus on yet

- Different DAS signatures for different propagation modes (sub-raleigh vs supershear)
- models based on loading rate to see if DAS data looks different
- Distinction of processes: Weak layer colapse, crown and staunchwall fractures
- Relation to snow properties in models
- relationship between crack propagation and release area

3 Methods

3.1 Wave Propagation Modeling

The model described in 3.2 is implemented using the finite-element (FE) method in SALVUS. To discretize a domain, the domain is divided into a finite number of elements (Zienkiewicz et al., 2025). The boundaries of these elements describe the mesh, the mesh has a finite number of grid-points (edge points of the elements) (Zienkiewicz et al., 2025). The governing equations are determined at every element node, so for rectangular elements with nodes 1,2,3 and 4, there is the system $r^1 = \begin{pmatrix} r_1^1 \\ r_2^1 \\ r_3^1 \\ r_4^1 \end{pmatrix}$ for force and $u^1 = \begin{pmatrix} u_1^1 \\ u_2^1 \\ u_3^1 \\ u_4^1 \end{pmatrix}$ for displacement (Zienkiewicz et al., 2025). The residual or characteristic relationship of linear elastic elements is of the form $r^1 = f^1 - K^1u^1$, where f are the nodal forces distributing load on the nodes and K is the stiffness matrix of a specific element (Zienkiewicz et al., 2025). Because in FE, the approximation of the wavefield by basis functions leads to a representation over the whole domain and not just at a set of points (Afanasiev et al., 2019). This means integrals can be calculated over surfaces or even volumes (Afanasiev et al., 2019).

The governing equation considered in this case is the elastic wave equation, because real-life crack propagation has P- and S-waves. The wave equation for a heterogenous elastic solid in the n-dimensional domain G is

$$\rho\delta_t^2 u = \nabla \cdot \sigma + f \tag{1}$$

(Afanasiev et al., 2019; Aki & Richards, 1981). Stress can be related to displacement using Hoek's law: $\sigma = \frac{C}{\nabla u}$, where C is a 4th order tensor which characterizes the stiffness of the medium (Afanasiev et al., 2019). Combining 1 with Hoek's law, this leads to

$$\rho\delta_t^2 u = \nabla \cdot \frac{C}{\nabla u} + f, \tag{2}$$

which is the strong form of the linear elastic wave equation.

Boundary conditions, such as free surfaces, as well as initial conditions must be stated explicitly, they have the form $\sigma \cdot \hat{n}|_{\delta_G} = \bar{\tau}(x), u|_{t=0} = \bar{u}(x), v|_{t=0} = \bar{v}(x)$ (Afanasiev et al., 2019). In this case, $\bar{\tau}(x), \bar{u}(x), \bar{v}(x)$ are the spatially variable distributions of traction, displacement and velocity (Afanasiev et al., 2019). The outward pointing normal vector on the boundary of domain G is described by $\hat{n}|_{\delta G}$.

FE methods use the weak form of the governing equations (Zienkiewicz et al., 2025). To obtain the weak form of the linear elastic wave equation in 1, the whole equation needs to be multiplied with a test function w, which is not time-dependent (Afanasiev et al., 2019). The weak form enforces the solution of the governing equation across the domain, instead of solving the equation everywhere, an integration over time leads to

$$\int_G \rho w \cdot \delta_t^2 u \, d^n x = \int_G w \cdot (\nabla \cdot \sigma) \, d^n x + \int_G w \cdot f \, d^n x \tag{3}$$

(Afanasiev et al., 2019). The divergence theorem relates the flux of a vector field through a closed surface to the divergence of the field in an enclosed volume: $\iiint_V (\nabla \cdot F) dV = \oiint_s (F \cdot \hat{n}) dS$, where F is the vector field. When this is applied to the first term on the right in 3, this leads to

$$\int_G \rho w \cdot \delta_t^2 u \, d^n x = \int_{\delta G} w \cdot (\sigma \cdot \hat{n}) \, d^{n-1} x - \int_G (\nabla w : \sigma) \, d^n x + \int_G w \cdot f \, d^n x \tag{4}$$

(Afanasiev et al., 2019). The first integral on the left side of 4 is the boundary integral, which changes depending on the boundary conditions implemented. The inegral on the right hand side is the mass term, the second on the left hand side is the stiffness term, and the last on the left hand side is the external forcing term, all in 4. Equation 4 can be compacted to

$$(\rho \delta_t^2 u, w)_G - \langle \sigma \cdot \hat{n}, w \rangle_{\delta G} + (\sigma, \nabla w)_G = (f, w)_G, \tag{5}$$

using the notations $(a, b)_G = \int_g a \cdot b d^n x$, $\langle a, b \rangle_{\delta_G} = \int_{\delta G} a \cdot b d^{n-1} x$ as well as Hoek's law (Afanasiev et al., 2019). This can be even more compacted by describing a mathematical model I with M for mass term, B for boundary term, S for stiffness and F for forcing term:

$$I_I(M) - I_I(B) + I_I(S) = I_I(F) = \sum_{P \in \{M,B,S,F\}} I_I(P) = 0 \tag{6}$$

(Afanasiev et al., 2019).

To solve 6 the Galerkin approach is used, the test function w and solution u are approximated by ϕ, which are a set of functions from the same space (Afanasiev et al., 2019). Any discrete function, for example displacement can be approximated this way:

$$g(x) \approx \sum_{j=1}^{N} F_j \phi_j(x), \tag{7}$$

where F_j is the expansion of the basis and g is the continous function which is the approximation (Afanasiev et al., 2019). The derivatives of a function can then be approximated by the derivatives of ϕ.

To evaluate the integrals in 6, numerical quadrature is used Afanasiev et al., 2019. A certain number of intergration weights W are combined, g(x) and the derivative of g(x) are evaluated at a set of integration points x_i (Afanasiev et al., 2019). This resluts in

$$\int_G g(x) d^n x \approx \sum_{i=1}^{M} W_i^G \sum_{j=1}^{N} F_j \phi_j(x_i) \tag{8}$$

(Afanasiev et al., 2019). In FE, an element-specific referency coordinate system θ is used. To transfrom the integrals and derivaties to physical coordinates x, the element-specific Jacobian $\frac{\delta x}{\delta \theta}$ is used (Afanasiev et al., 2019). Equations 7 and 8 and their derivatives become:

$$g(x) \approx \sum_{N}^{j=1} F_j \phi_j(x) \tag{9}$$

(Afanasiev et al., 2019). In equation 9, the derivatuves of g(x) can be approximated by ϕ_j, the derivatives of the set of functions approximating the test function w and the solution u (Afanasiev et al., 2019). This results in:

$$\nabla g(x) \approx \sum_{j=1}^{N} F_j \nabla \phi_j(x). \tag{10}$$

In equations 9 and 10, the function space V is approximated as $V_h \approx V$, this means the accuracy of the approximation in equation 10 is dependent on how well the original function space is approximated (Afanasiev et al., 2019). The integrals of displacement in respect to $(u, w)_G, (\nabla u \cdot \hat{n}, w)_{\delta G}$ are evaluated using numerical quadrature, where a set of M integration weights W_i are combined with g(x) and $\nabla g(x)$ are evaluated at a set of points x_i (Afanasiev et al., 2019).

3.2 Cohesive Zone Model

The rupture model used in the simulations is implemented as a mixed-mode Cohesive Zone Model (CZM) following Mahajan and Joshi, 2008. This model was chosen because there is a narrow band ahead of the fracture zone, the cohesive zone, where the material is already weakened. This means that before a point in the material cracks, there is already weakening of the material before. This seems to be reasonable for representing crack propagation in snow, since failure occurs as a consequence of loading and is not spontaneous. The computational domain extends 400 m in the horizontal (x) direction and 3 m in the vertical (y) direction and contains a 1.5 m snow layer over a 1.5 m air layer. Material parameters for the snow layer are taken from Guillemot et al., 2021. The mixed-mode behavior is implemented because in reality, crack propagation can involve both normal and shear displacement.

The mesh is generated to resolve frequencies up to the 95th percentile of the source wavelet's frequency content. Cohesive tractions are prescribed on zero-thickness interface elements and depend on the normalized opening $\xi = \frac{\delta}{\delta_c}$. The zero-thickness interface elements are implemented by not solving for volumetric strain but displacement between interfaces instead. For the linear softening law used here the traction in a given cohesive element follows

$$T(\xi) = T_{\text{peak}}\,(1 - \xi) \quad \text{for } 0 \le \xi \le 1, \qquad T(\xi) = 0 \text{ for } \xi > 1,$$

where T_{peak} is the peak cohesive traction and δ_c the critical opening. The normalized opening can be related to the local crack coordinate as $\xi = \frac{(x-x_0)}{l_{crack}}$ for a crack front with initial position x_0 and local process length l_{crack}.

The mixed-mode potential used in the code follows the form from Mahajan and Joshi, 2008. With $\Delta = (\Delta_n, \Delta_t)$ denoting normal and tangential separations, the potential is written as

$$\phi(\Delta) = \phi_n \exp\left(-\frac{\Delta_n}{\delta_n}\right)\left\{\left[1 + \frac{\Delta_n}{\delta_n}\right] + q\left[1 - \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right)\right]\right\},$$

from which the normal and tangential tractions are obtained, the implementation uses the same closed-form expressions. This is so that peak tractions occur at $\Delta_n = \delta_n$ and $|\Delta_t| = \delta_t/\sqrt{2}$. The normal and shear works of separation relate to peak strengths via

$$\phi_n = e\,\sigma_{\max}\,\delta_n, \qquad \phi_t = \sqrt{\tfrac{e}{2}}\,\tau_{\max}\,\delta_t,$$

and the critical openings for pure modes are computed as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \qquad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

with the mixed-mode critical opening

$$\delta_c = \sqrt{(1 - r_{\text{mode}})\,\delta_{n,c}^2 + r_{\text{mode}}\,\delta_{s,c}^2}\,.$$

Lamé parameters are derived from input seismic speeds and density: $v_s = \sqrt{\frac{\mu}{\rho}}$ so that $\mu = \rho v_s^2$, and $v_p = \sqrt{\frac{(\lambda+2\mu)}{\rho}}$ so that $\lambda = \rho(v_p^2 - 2v_s^2)$. Young's modulus is then computed from the Lamé parameters: $E = \dfrac{\mu(3\lambda + 2\mu)}{\lambda + \mu}$ (Aki & Richards, 1981). These conversions are applied for every material in the model.

The critical opening distances for pure normal (Mode I) and shear (Mode II) separations are computed from the corresponding fracture energies, strenghts, and peak tractions as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \qquad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

and combined for mixed-mode behaviour using

$$\delta_c = \sqrt{(1 - r_{\mathrm{mode}})\, \delta_{n,c}^2 + r_{\mathrm{mode}}\, \delta_{s,c}^2},$$

where $r_{\mathrm{mode}} \in [0, 1]$ controls the mode I/II mix, where 0 is pure mode I and 1 is pure mode II. In the current implementation the slip on each subfault follows the prescribed linear slip law up to the critical slip distance. The mode mix is an user-input parameter, while all the other parameters are derived from laboratory experiments from Mahajan and Joshi, 2008.

The characteristic Irwin length for shear is calculated as $l_{ch} = \frac{E_{\mathrm{eff}} G_{II}}{\tau_s^2}$, and the cohesive zone length is taken as $l_{cz} = \frac{\pi}{8} l_{ch}$, which sets the spatial extent over which cohesive weakening is active ahead of the crack tip.

Rupture propagation speed is chosen with respect to the local shear-wave speed and the bounds discussed in Trottet et al., 2022. Where appropriate, the rupture speed is corrected for finite slab thickness and bending stiffness to account for slab flexure effects. The rupture front is discretized into a sequence of subfaults: each subfault is activated at an onset time

$$t_o = \frac{x - x_{\mathrm{start}}}{v_{\mathrm{rupture}}},$$

using the subfault horizontal position x, the rupture starting position x_{start}, and the prescribed rupture speed v_{rupture}.

Each activated subfault is represented as a point/source in the SALVUS source array. The scalar seismic moment for a subfault is computed from the shear modulus, the prescribed slip, and the subfault width Δx:

$$M_0 = \mu\, x_{\mathrm{slip}}\, \Delta x.$$

Moment-tensor components are set according to the user-specified mode-mix ratio, e.g. here the implementation uses $m_{xx} = 0,\ m_{xy} = M_0\, r_{\mathrm{mode}},\ m_{yy} = -(1 - r_{\mathrm{mode}}) M_0$, which yields pure mode behaviour at the extremes of $r_{\mathrm{mode}} = 0$ or 1, which means that an additional component of the moment tensor source becomes 0. All subfault sources use the same central frequency, wavelet shape, and amplitude scaling so the rupture is represented as a coherent moving source in the wave propagation simulations.

The following numerical settings and checks are applied consistently across simulations to ensure stability and reproducibility:

- Mesh resolution: element/subfault spacing h is chosen to resolve the cohesive zone and the shortest seismic wavelength; practical rule: $h \leq \min(l_{cz}/4,\ \lambda_{\min}/8)$, where $\lambda_{\min} = v_{\min}/f_{95}$ and f_{95} is the 95%ile source frequency.
- Time stepping: explicit scheme with CFL-like stability criterion $\Delta t \lesssim s\, h/v_{\max}$ and safety factor $s \approx 0.3$–0.5.
- Boundaries: use absorbing boundary conditions to minimize reflections at domain edges.
- Rupture discretization: subfault onset times follow $t_o = (x - x_{\mathrm{start}})/v_{\mathrm{rupture}}$; rise time is taken as $t_{rise} \approx l_{cz}/v_{\mathrm{rupture}}$; scalar moment per subfault $M_0 = \mu\, x_{\mathrm{slip}}\, \Delta x$.

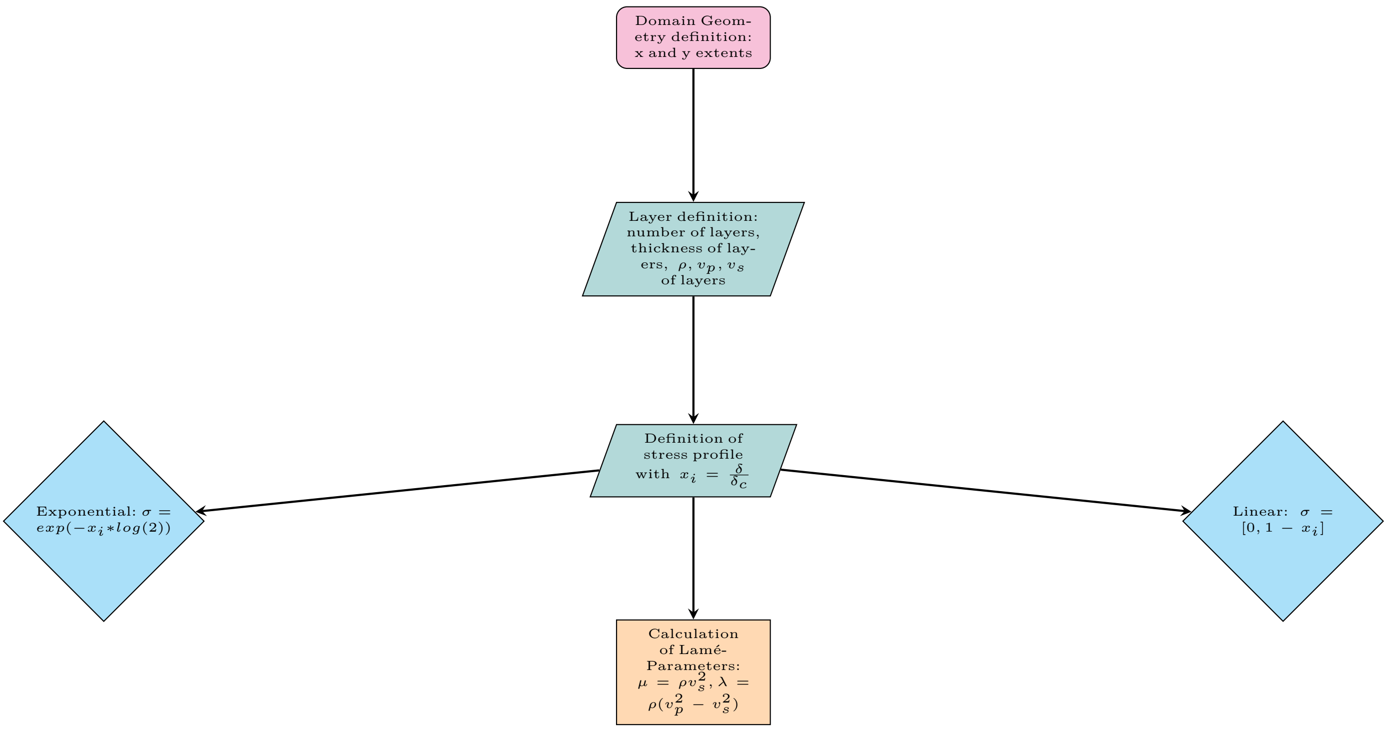
Key parameters and their provenance used in the simulations are summarised below:

Parameter	Typical value / source
σ_n	0.00184 MPa from Mahajan and Joshi, 2008
τ_s	0.004 MPa from Mahajan and Joshi, 2008
G_I, G_{II}	0.35, 0.50 $J/m^s$from Mahajan and Joshi, 2008
v_s, v_p, ρ	150, 332 m/s from Guillemot et al., 2021 (converted to μ, λ, E)
r_{mode}	User input (0 = Mode I, 1 = Mode II)

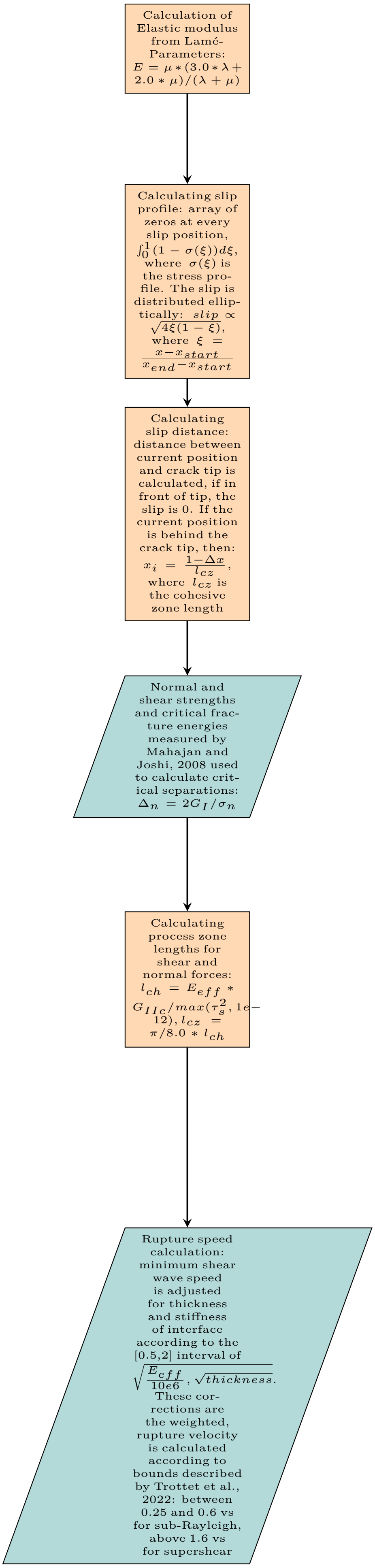
NOTE: add to text if this actually works out parameters τ_s, and σ_n, were scaled by the same factor. This results in scaling of cohesive zone length and critical slip distance. This was done so that the slip and cohesive zone length can be resolved by the mesh, without increasing computation or memory requirements.

3.3 Code workflow

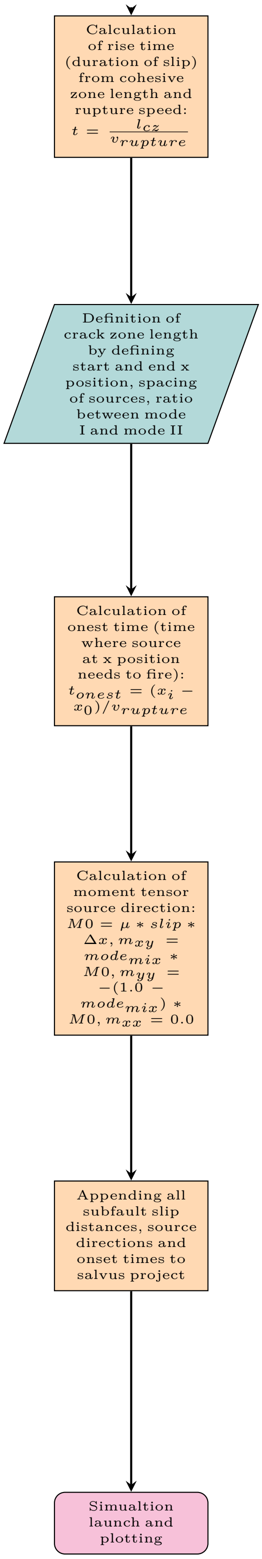
Part 1: Domain and Stress Profile Setup



Part 2: Material Properties and Process Zone Calculation



Part 3: Rupture Propagation and Source Definition



4 Notes

- Look into wave modes (flexural and other mode)

5 Bibliography

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